Intraday Volume Lab

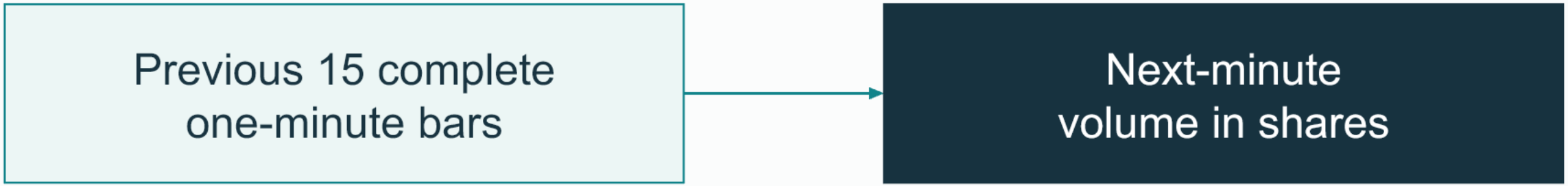
Next-minute volume forecasting

Bikash Kushwaha

Python research portfolio
Synthetic demonstration with fictional instruments

Forecasting objective

Predict interval volume in shares from information already available.



Causal windows

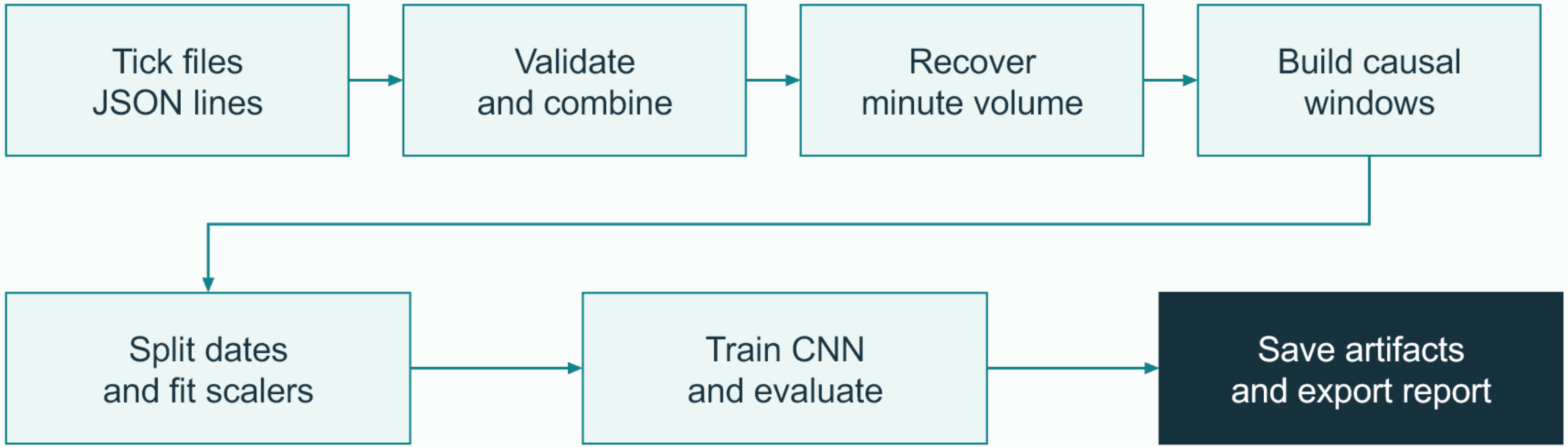
A window stays inside one instrument and one local session.

Training target

log1p(shares), standardized using training data only.

Project architecture

One shared Python module supports the notebook and command line.



Python / TensorFlow / pandas / scikit-learn / Matplotlib / Jupyter / pytest / Ruff

Dataset and chronological evaluation

Synthetic data: ALPHA and BETA across eight short sessions.

1,936
tick records

960
complete minute bars

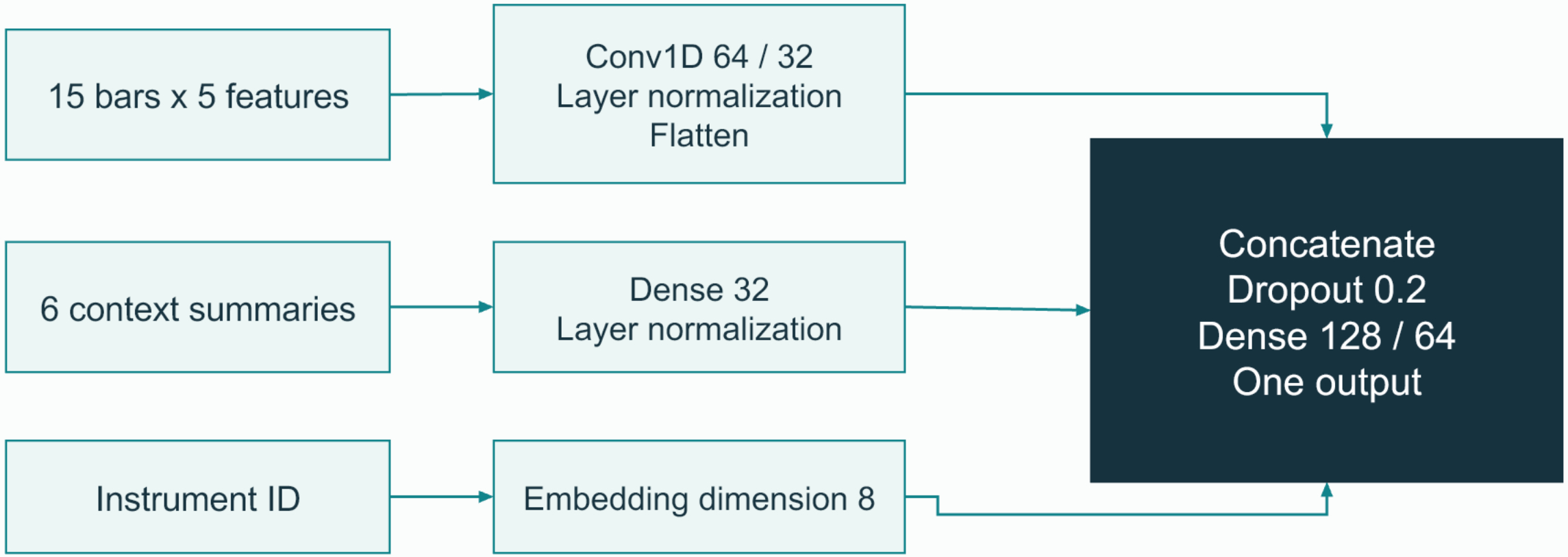
720
forecasting windows

Partition	Dates in January 2026	Session dates	Windows
Training	05-09	5	450
Validation	12	1	90
Test	13-14	2	180

Explicit timezone and units. Daily resets and feed gaps handled before window creation.

Model and input features

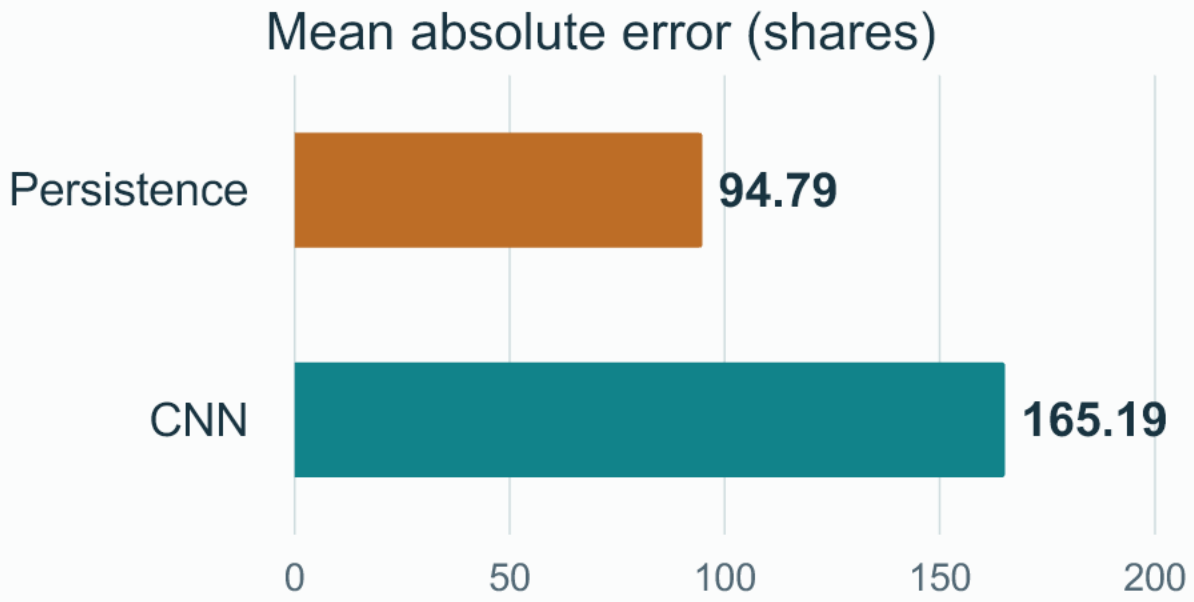
Three input branches feed a single volume regression output.



Adam learning rate 0.0005. MSE loss. Fixed seed 42.

Persistence performs better in this run

Held-out synthetic test set: 180 windows after five training epochs.



Metric	CNN	Persistence
MAE	165.19	94.79
RMSE	220.78	128.84
R2	0.619	0.870

This short demonstration verifies the workflow.
It does not establish real-market forecasting performance.

Testing and limitations

30 passing tests

Interval volume and daily resets

Causal windows and date boundaries

Scalers fitted on training only

Model save and reload

Research boundaries

Eight synthetic dates

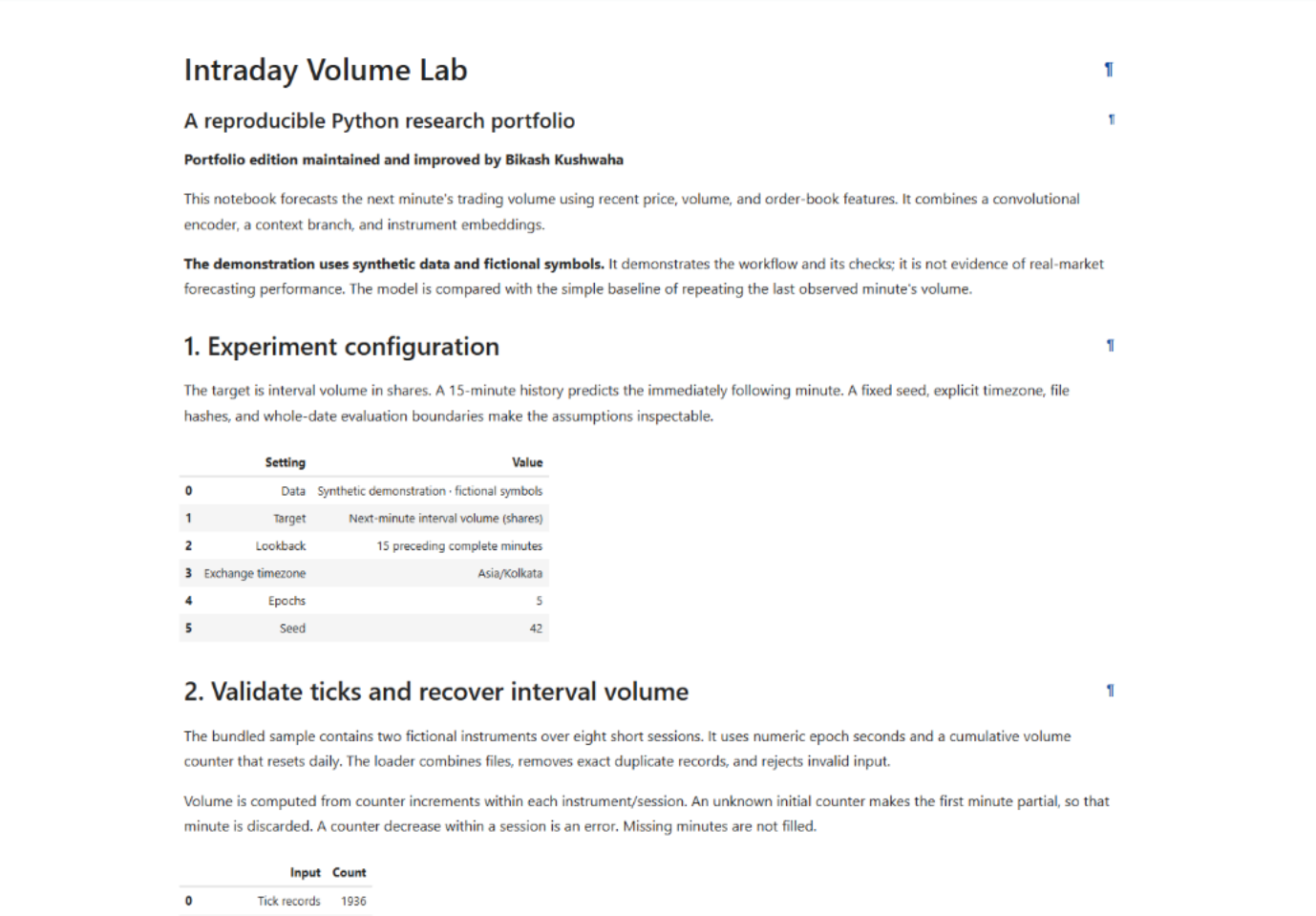
Five training epochs

No real-market dataset

No live inference or CMEM estimator

Windows CPU and Ubuntu CI passed. Ruff checks formatting and imports.

Run the project and explore the report



Intraday Volume Lab ¶

A reproducible Python research portfolio ¶

Portfolio edition maintained and improved by Bikash Kushwaha

This notebook forecasts the next minute's trading volume using recent price, volume, and order-book features. It combines a convolutional encoder, a context branch, and instrument embeddings.

The demonstration uses synthetic data and fictional symbols. It demonstrates the workflow and its checks; it is not evidence of real-market forecasting performance. The model is compared with the simple baseline of repeating the last observed minute's volume.

1. Experiment configuration ¶

The target is interval volume in shares. A 15-minute history predicts the immediately following minute. A fixed seed, explicit timezone, file hashes, and whole-date evaluation boundaries make the assumptions inspectable.

	Setting	Value
0	Data	Synthetic demonstration · fictional symbols
1	Target	Next-minute interval volume (shares)
2	Lookback	15 preceding complete minutes
3	Exchange timezone	Asia/Kolkata
4	Epochs	5
5	Seed	42

2. Validate ticks and recover interval volume ¶

The bundled sample contains two fictional instruments over eight short sessions. It uses numeric epoch seconds and a cumulative volume counter that resets daily. The loader combines files, removes exact duplicate records, and rejects invalid input.

Volume is computed from counter increments within each instrument/session. An unknown initial counter makes the first minute partial, so that minute is discarded. A counter decrease within a session is an error. Missing minutes are not filled.

	Input	Count
0	Tick records	1936

Local workflow

Use Python 3.12.
Create .venv and install requirements-lock.txt.
Run the tests and notebook.
Export the static report.

GitHub repository

BikashCoder31/intraday-volume-lab

intraday-volume-lab.vercel.app

Complete commands and data policy are in the README.